LINEAR LEAST-SQUARES REGRESSION

- Before describing least-squares regression, we will first review some basic concepts from the field of statistics. These include the mean, standard deviation, residual sum of the squares, and the normal distribution.
- Assuming you are familiar with these subjects, you may skip...

Arithmetic Mean ($\bar{y}$)

- The arithmetic mean ($\bar{y}$) of a sample is defined as the sum of the individual data points (y_i) divided by the number of points (n), or

$$\bar{y} = \frac{\sum y_i}{n}$$

- where the summation (and all the succeeding summations in this section) is from i = 1 through n.

Measures of Spread

- The simplest measure of spread is the range, the difference between the largest and the smallest value. The most common measure of spread for a sample is the standard deviation (s_y) about the mean

$$s_y = \sqrt{\frac{S_t}{n-1}}$$

where S_t is the total sum of the squares of the residuals between the data points and the mean, or $S_t = \sum(y_i - \bar{y})^2$

- The spread can also be represented by the square of the standard deviation, which is called the variance:

$$s_y^2 = \frac{\sum(y_i - \bar{y})^2}{n-1}$$

Measures Variance

- We should note that an alternative, more convenient formula is available to compute the variance:

$$s_y^2 = \frac{\sum y_i^2 - \frac{\left(\sum y_i\right)^2}{n}}{n-1}$$

This version does not require precomputation of $\bar{y}$ and yields an identical result

Measures Variance

- A final statistic that has utility in quantifying the spread of data is the coefficient of variation (c.v.). This statistic is the ratio of the standard deviation to the mean. It is often multiplied by 100 so that it can be expressed in the form of a percent:

$$c.v. = \frac{s_y}{\bar{y}} \times 100\%$$

LINEAR LEAST-SQUARES REGRESSION

- Where substantial error is associated with data, the best curve-fitting strategy is to derive an approximating function that fits the shape or general trend of the data without necessarily matching the individual points.
- One approach to do this is to visually inspect the plotted data and then sketch a “best” line through the points.
- One way to do this is to derive a curve that minimizes the discrepancy between the data points and the curve.

LINEAR LEAST-SQUARES REGRESSION

- The simplest example is fitting a straight line to a set of paired observations: (x_1, y_1), (x_2, y_2), ... … , (x_n, y_n).
- The mathematical expression for the straight line is

$$y = a_0 + a_1 x + e$$

- where a_0 and a_1 are coefficients representing the *intercept* and the *slope*, respectively, and e is the error, or residual, between the model and the observations, which can be represented by rearranging above equation as

$$e = y - a_0 - a_1 x$$

Criteria for a "Best" Fit

- One strategy for fitting a "best" line through the data would be to minimize the sum of the residual errors for all the available data, as in

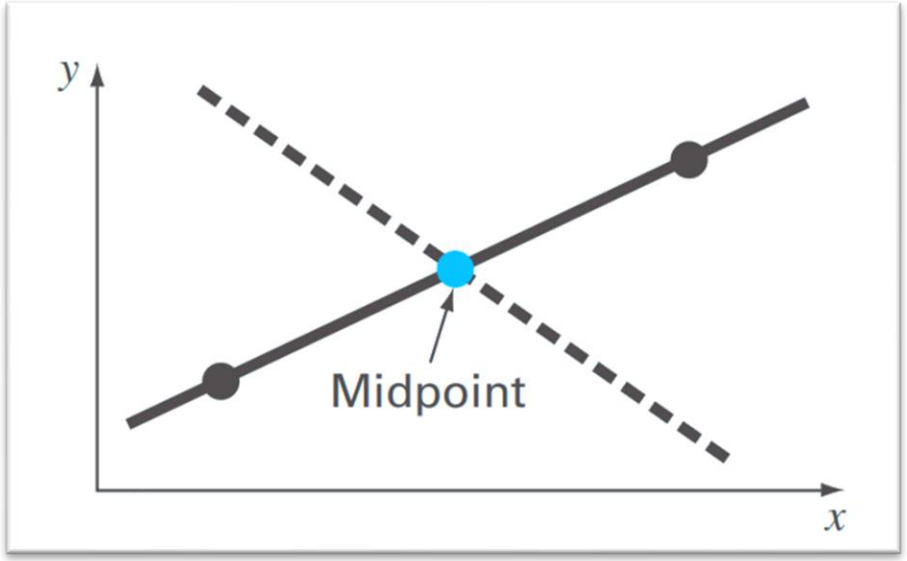
$$\sum_{i=1}^{n} e_i = \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)$$

- Obviously, the best fit is the line connecting the points. However, any straight line passing through the midpoint of the connecting line (except a perfectly vertical line) results in a minimum value of the above equation equal to zero because positive and negative errors cancel.

Criteria for a "Best" Fit

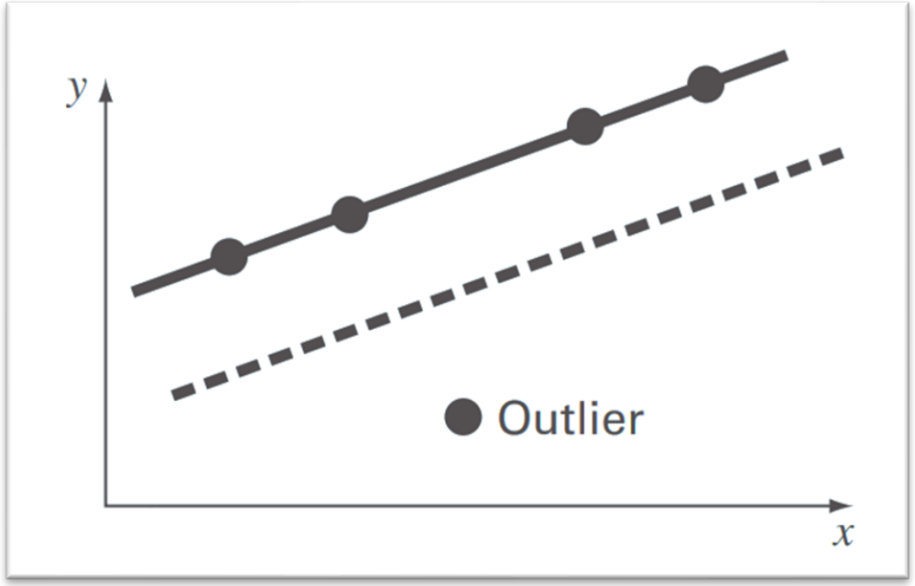
- One way to remove the effect of the signs might be to minimize the sum of the absolute values of the discrepancies, as in

$$\sum_{i=1}^{n} |e_i| = \sum_{i=1}^{n} |(y_i - a_0 - a_1 x_i)|$$



Minimax Criterion

- A third strategy for fitting a best line is the minimax criterion. In this technique, the line is chosen that minimizes the maximum distance that an individual point falls from the line. As depicted in Fig.
- This strategy is ill-suited for regression because it gives undue influence to an outlier—that is, a single point with a large error.



Least-Squares

- It should be noted that the minimax principle is sometimes well-suited for fitting a simple function to a complicated function
- A strategy that overcomes the shortcomings of the afore mentioned approaches is to minimize the sum of the squares of the residuals:

$$S_r = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)^2$$

- This criterion, which is called least squares, has a number of advantages, including that it yields a unique line for a given set of data.

Least-Squares Fit of a Straight Line

- To determine values for a_0 and a_1, in the previous equation is differentiated with respect to each unknown coefficient:

$$\frac{\partial S_r}{\partial a_0} = -2\sum(y_i - a_0 - a_1 x_i)$$

$$\frac{\partial S_r}{\partial a_1} = -2\sum[(y_i - a_0 - a_1 x_i)x_i]$$

- Note that we have simplified the summation symbols; unless otherwise indicated, all summations are from i = 1 to n.

Least-Squares Fit of a Straight Line

- Setting these derivatives equal to zero will result in a minimum S_r
- If this is done, the equations can be expressed as

$$0 = \sum y_i - \sum a_0 - \sum a_1 x_i$$

$$0 = \sum x_i y_i - \sum a_0 x_i - \sum a_1 x_i^2$$

Least-Squares Fit of a Straight Line

Now, realizing that $\sum a_0 = na_0$, we can express the equations as a set of two simultaneous linear equations with two unknowns (a_0 and a_1):

$$\sum y_i = na_0 + \left(\sum x_i\right) a_1$$

$$\sum x_i y_i = \left(\sum x_i\right) a_0 + \left(\sum x_i^2\right) a_1$$

These are called the normal equations.

Least-Squares Fit of a Straight Line

They can be solved simultaneously for

$$a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$$

$$a_0 = \bar{y} - a_1\bar{x}$$

where $\bar{y}$ and $\bar{x}$ are the means of y and x, respectively.

Example of Least-Squares Fitting

Fit a straight line to the values given in the following table

i	x_i	y_i
1	10	25
2	20	70
3	30	380
4	40	550
5	50	610
6	60	1,220
7	70	830
8	80	1,450

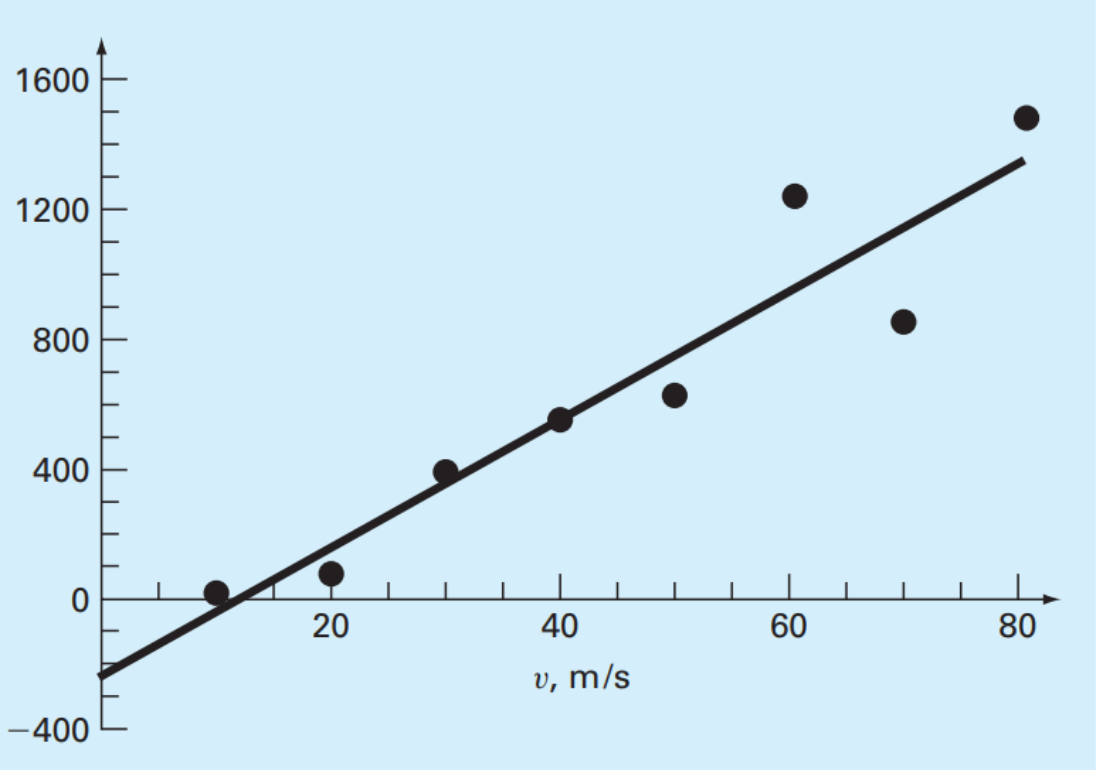
Example of Least-Squares Fitting

i	x_i	y_i	x_i^2	$x_i y_i$
1	10	25	100	250
2	20	70	400	1,400
3	30	380	900	11,400
4	40	550	1,600	22,000
5	50	610	2,500	30,500
6	60	1,220	3,600	73,200
7	70	830	4,900	58,100
8	80	1,450	6,400	116,000
$\sum$	360	5,135	20,400	312,850

The means can be computed as

$\bar{x} = 360/8 = 45$, $\bar{y} = 5{,}135/8 = 641.875$

- The slope and the intercept can then be calculated as,

$$a_1 = \frac{8(312{,}850) - 360(5{,}135)}{8(20{,}400) - (360)2} = 19.47024$$

$$a_0 = 641.875 - 19.47024(45) = -234.2857$$